Jason Tang

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Education

University of Toronto Sept. 2023 – May 2028
BASc, Computer Engineering | GPA: 3.66/4.00

- **Relevant Coursework:** Data Structures & Algorithms, Operating Systems, Computer Networking, Digital Systems
- **Awards:** Edward S Rogers Sr. Scholarship, Faculty of Applied Science and Engineering Scholarship
- **Activities:** Tri-Campus Volleyball Captain; Badminton Team; Ultimate Frisbee

Experience

AMD May 2025 – Aug. 2026
Software Engineer Intern *Toronto, ON*

- Developed and scaled an internal regression-management platform serving **1,000+ engineers** and processing **3.1M tests** monthly, building full-stack features across Vue, TypeScript, Node.js/Express, Ruby, and PostgreSQL
- Reduced failure triage to **30 seconds** from 15–30 minutes by building an **AI debugging agent** that correlated logs, scheduler events, and regression metadata to automatically identify the likely root causes of regression failures
- Improved reliability for **239K jobs/month** by building job lifecycle services that reconcile job state after failures
- Enabled **10K+ monthly AI operations** with an authenticated **TypeScript MCP** integration used by 100+ engineers
- Saved **20+ hours weekly** and cut release cycles by **30%** via deployment automation in Python, Ruby, and Jenkins
- Secured traceability for **1.4M+ regression runs** with validation workflows for a 113-repo Perforce-to-Git migration

U+ Education June 2023 – Sept. 2023
Web Developer *Ottawa, ON*

- Designed and launched two responsive WordPress websites serving **100+ customers** with reusable page layouts
- Improved page-load speed by **30%** through image optimization, caching, and lazy loading; grew organic traffic by **50%+** through technical SEO and analytics-driven content updates across both websites
- Spearheaded the company's **expansion into Ottawa**, launching its local web presence and student outreach

Digitera Interactive June 2020 – Sept. 2020
Full-stack Developer Intern *Ottawa, ON*

- Built an event-planning app with scheduling, registration, and accounts using HTML, CSS, JavaScript, and MySQL
- Designed the database schema and implemented form validation, filtering, and error handling for CRUD operations
- Implemented authentication and **role-based permissions** to secure event management and registration features

Projects

LiftLens | *Python, PyTorch, TypeScript, Next.js, ONNX Runtime, FastAPI, Redis* Oct. 2026 – Present

- Built a **computer vision** app using **pose estimation** and bar tracking to analyze joint angles in Olympic lifts
- Fine-tuned **YOLOv8** on custom-labeled frames to detect barbells, applied **Kalman filtering** to stabilize tracking through motion blur and occlusion, and estimated bar velocity using the 450 mm plate diameter and frame timing
- Ran models in-browser with **ONNX Runtime Web** and **TypeScript** for live webcam inference entirely on the client
- Built a **Next.js** app with a **Redis** job queue for server-side processing of uploaded videos and rep comparison

Mini ML Framework | *Rust, CUDA, TypeScript, PyTorch, TensorFlow* Aug. 2026

- Built a Rust ML framework with tensors, reverse-mode autodiff, CPU/CUDA backends, and TypeScript bindings
- Sped up training **3.2×** over CPU by writing custom CUDA kernels for elementwise operations and reductions
- Verified tensor ops, gradients, and optimizers against PyTorch and TensorFlow with cross-framework tests
- Trained a 235K-parameter MLP on MNIST to **97.4%** test accuracy using cross-entropy loss and the Adam optimizer

Technical Skills

Languages: Python, TypeScript, JavaScript, Rust, C++, C, Ruby, SQL, SystemVerilog, Verilog, C#, Perl, HTML/CSS

Frameworks & Tools: React, Next.js, Vue.js, Node.js, Express.js, FastAPI, PyTorch, TensorFlow, ONNX Runtime, OpenCV, Unity, Linux, Docker, Kubernetes, PostgreSQL, Redis, RabbitMQ, Jenkins, Git, REST APIs, MCP, CUDA, AWS

Concepts: Data Structures & Algorithms, OOP, Concurrency, Distributed Job Scheduling, Event-Driven Architecture, WebSockets, CI/CD, Unit Testing, Machine Learning, Neural Networks, Computer Vision, RAG, AI Agents, Agile